

Paper IV — Compact-Object Gate of the BPB/MBE Framework

EOS-Coherent Tidal Suppression in Neutron Stars from the Strict Dictionary

v0.4.1 HAL corpus release

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Abstract

We extend the BPB/MBE programme to compact objects by propagating the strict dictionary into neutron-star structure. This is the fourth independent gate of the current branch: after the low-redshift/growth gate, the SPARC galactic sign gate, and the CMB/Weyl strict face-minus gate, we test whether the same dictionary survives dense nuclear matter. The strict neutron-star coupling is fixed to $\kappa_{\text{strict}} = 0.965543842$, inherited from the Paper-0/Paper-III strict dictionary closure rather than tuned on neutron-star data. We run eight nuclear equations of state: APR, BSk24, DD2, DDME2, NL3wrL55, SFHo, SLy4, and TW, using standard relativistic and nonrelativistic dense-matter EOS families as an eight-model stress test [1, 2, 3, 4, 5, 6, 7, 8]. All eight EOSs pass the strict checkpoint gate. The scanned mass-radius sector remains viable, with sequence maxima between $2.043M_{\odot}$ and $2.753M_{\odot}$. The tidal response is coherently suppressed. Across the full finite checkpointed sequences, the post-F source lock gives $\langle \Delta\Lambda/\Lambda \rangle_{\text{seq}} = -1.0456\%$ with inter-EOS scatter 0.082 percentage point. At the canonical mass $1.4M_{\odot}$, all eight EOSs give $\Delta\Lambda/\Lambda(1.4M_{\odot}) < 0$, with mean -2.7019% , range $[-2.982\%, -2.385\%]$, and scatter 0.213 percentage point. Version v0.4.1 incorporates the post-F no-minus tidal-convention source lock and keeps the interpretation deliberately bounded: the exported products support an effective strict-dictionary tidal/Love-sector response at fixed exported compactness, not yet a native scalar-tensor Love-number solver and not a full gravitational-wave population likelihood. Paper IV is therefore the compact-object survival gate of the BPB/MBE corpus, with a signed, percent-level, EOS-coherent prediction.

1 Executive claim

The compact-object result can be stated in one sentence:

The strict BPB/MBE dictionary survives an independent eight-EOS neutron-star gate and predicts a coherent tidal suppression: about -1.0% over full sequences and -2.70% at $1.4M_{\odot}$, with all eight EOSs negative.

This is deliberately not phrased as a discovery claim from gravitational-wave data. It is a structural prediction and consistency gate. The purpose of Paper IV is to show that the branch selected by Papers I–III does not collapse when pushed into neutron-star interiors.

BPB/MBE gate sequence leading to the compact-object test

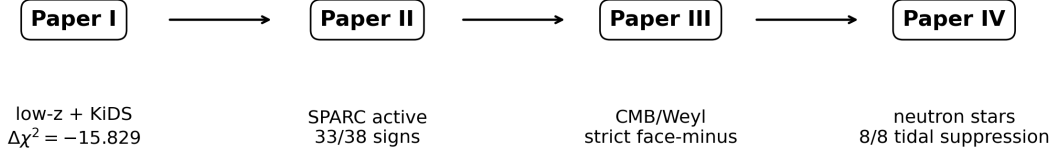


Figure 1: Narrative position of Paper IV in the current BPB/MBE sequence. The neutron-star result is not an isolated calculation; it is the compact-object continuation of the branch already selected by the low-redshift, galactic, and CMB/Weyl gates.

2 Corpus placement and release navigation

Paper IV is the compact-object member of the BPB/MBE corpus. Paper 0 defines the strict dictionary; Paper I tests the low-redshift/growth continuation; Paper II tests the galactic/SPARC continuation; Paper III tests the CMB/Weyl continuation; and the present paper tests whether the same branch survives dense nuclear matter. The release-level navigation is carried by the corpus artefacts `00_README_CORPUS`, `00_MANIFEST_CORPUS`, `00_CLAIMS_INDEX`, and `00_REVIEWER_ROUTE`. Those files are included in the source bundle under `release_index/`.

The Paper-IV claim is intentionally narrower than a gravitational-wave discovery claim. It is a structural gate: the strict branch must give finite neutron-star sequences, must keep the maximum masses above the observed two-solar-mass scale for the EOS set tested [11, 12], and must produce a signed, EOS-coherent tidal response without tuning neutron-star data.

3 Dictionary continuity

3.1 Why the strict branch is the main branch

The present neutron-star branch does not revive the older large-amplitude neutron-star dictionary as the main result. The main branch is the strict dictionary selected after the CMB/Weyl analysis. The formal closure file gives the following dictionary-level relations. Here the coupling entering the neutron-star split-relic closure is denoted β_{cl} , to avoid confusion with the low-redshift MAP1782 growth parameter $\beta_{\text{grow}}^{\text{MAP1782}} = -0.2106621406$. The closure JSON uses $\beta_{\text{cl}} = -0.12489553$, and therefore

$$Z_{\beta} = s_{\text{surv},0} = 0.7956924436, \quad (1)$$

$$\alpha_{\infty} = \frac{|\beta_{\text{cl}}|}{s_{\text{surv},0}} = \frac{0.12489553}{0.7956924436} = 0.1569645797, \quad (2)$$

$$S'(\phi_{\infty}) = 1 - s_{\text{surv},0} = 0.2043075564. \quad (3)$$

This correction is purely notational: $\alpha_{\infty} = 0.1569645797$ is the value recorded in the dictionary-closure JSON. It should not be recomputed using the MAP1782 low-redshift growth coefficient. Physically, $\beta_{\text{grow}}^{\text{MAP1782}}$ is a late-time cosmological growth-response coefficient, whereas β_{cl} is the compact-sector scalar normalization used to export the strict dictionary into the neutron-star tidal calculation. They live in different descendant projections of the same Paper-0 dictionary and are therefore kept notationally distinct. With this closure input, the strict neutron-star branch gives

$$\kappa_{\text{strict}} = 0.965543842. \quad (4)$$

Relative to the BSk24 reference threshold $\kappa_{\text{crit}} \simeq 0.89$, this gives

$$\kappa_{\text{strict}} - \kappa_{\text{crit}} = 0.075544, \quad \kappa_{\text{strict}}/\kappa_{\text{crit}} = 1.084881. \quad (5)$$

The analytic derivation of the bridge value remains a caveat, but the strict dictionary itself is fixed before the neutron-star run. In other words, the neutron-star sector is a test, not a fit.

3.2 Nearby branches as diagnostics only

The legacy and hybrid branches are kept only to check local stability. If the sign disappeared under tiny changes of κ , the result would be fragile. Instead, the triad diagnostic shows that nearby dictionaries preserve the sign and order of magnitude. This supports the robustness of the neutron-star response while keeping the strict dictionary as the only branch promoted to the main claim.

4 Numerical definition of the exported tidal response

4.1 Checkpointed sequence rows

Each strict checkpoint row stores, for a given EOS and central density, the stellar mass M , radius R , compactness proxy $2M/R$, topology flags, and a tidal block. The global gate summary counts finite rows used for the sequence-level statistics. Across the eight EOSs the gate uses 9436 finite rows and contains 75 topology-positive points.

4.2 Tidal block and current decomposition status

The tidal block stores a GR Love-number baseline and a strict-dictionary corrected tidal deformability. The GR baseline follows the standard relativistic tidal-deformability convention [9, 10]. It obeys

$$\Lambda_{\text{GR}} = \frac{2}{3} k_{2,\text{GR}} C^{-5}, \quad C = \frac{M}{R}. \quad (6)$$

The exported strict value can be written equivalently as

$$\Lambda_{\text{BPB}} = \frac{2}{3} k_{2,\text{eff}}^{\text{BPB}} C^{-5}, \quad (7)$$

where the compactness C is the exported compactness of the row. Thus the current fractional correction satisfies

$$\frac{\Lambda_{\text{BPB}} - \Lambda_{\text{GR}}}{\Lambda_{\text{GR}}} = \frac{k_{2,\text{eff}}^{\text{BPB}} - k_{2,\text{GR}}}{k_{2,\text{GR}}} \quad \text{within the exported row-level diagnostic.} \quad (8)$$

This remains an important v0.4 guardrail. The present outputs provide a robust effective tidal/Love-sector suppression at fixed exported compactness. They do not yet provide the full pair of GR and BPB radii and Love numbers needed to decompose the result into independent

$$\frac{\Delta\Lambda}{\Lambda} = \frac{\Delta k_2}{k_2} - 5 \frac{\Delta C}{C} = \frac{\Delta k_2}{k_2} + 5 \left(\frac{\Delta R}{R} - \frac{\Delta M}{M} \right) \quad (9)$$

terms. That full structural decomposition is therefore not claimed in v0.4. It remains the next native-solver/export task.

4.3 Post-F tidal-convention source lock

Version v0.4.1 uses the post-F source lock. The tidal convention is the no-minus-F convention patched in `src/ns/tidal_st.py`; the strict checkpoints used by the paper are post-patch; and the canonical-mass refinement is downstream of those checkpoints. The allowed claim is therefore the strict-dictionary effective tidal response recorded in the exported products. The forbidden claim remains a completed native scalar-tensor Love-number solver with exterior scalar matching.

The post-F canonical values supersede the older displayed-Lambda values from the v0.3.1 source. The fractional source-of-truth estimator is unchanged in sign and summary scale: eight out of eight EOSs remain negative at $1.4M_{\odot}$, with mean -2.701932% .

5 Global eight-EOS strict gate

Table 1 gives the global checkpoint summary. The decisive features are: all EOSs pass, all sequence means are negative, and all scanned maxima remain above $2M_{\odot}$.

Table 1: Strict global checkpoint gate. The “finite rows” column is the row count used by the global gate summary, not necessarily the total raw checkpoint row count.

EOS	finite rows	topology+	$M_{\min}-M_{\max}^{\text{seq}} [M_{\odot}]$	$M_{\max}^{\text{seq}} [M_{\odot}]$	mean $\Delta\Lambda/\Lambda$ [%]
APR	1308	9	0.660–2.204	2.204	-1.063
BSk24	1255	9	1.116–2.277	2.277	-0.996
DD2	1244	9	1.155–2.404	2.404	-1.025
DDME2	1155	10	1.174–2.471	2.471	-0.898
NL3wrL55	986	10	0.897–2.753	2.753	-1.065
SFHo	1139	8	1.088–2.054	2.054	-1.073
SLy4	1136	10	0.991–2.043	2.043	-1.210
TW	1213	10	1.220–2.052	2.052	-1.035
Total/mean	9436	75	–	–	-1.046
Scatter	–	–	–	–	0.082

Eight-EOS strict global checkpoint gate

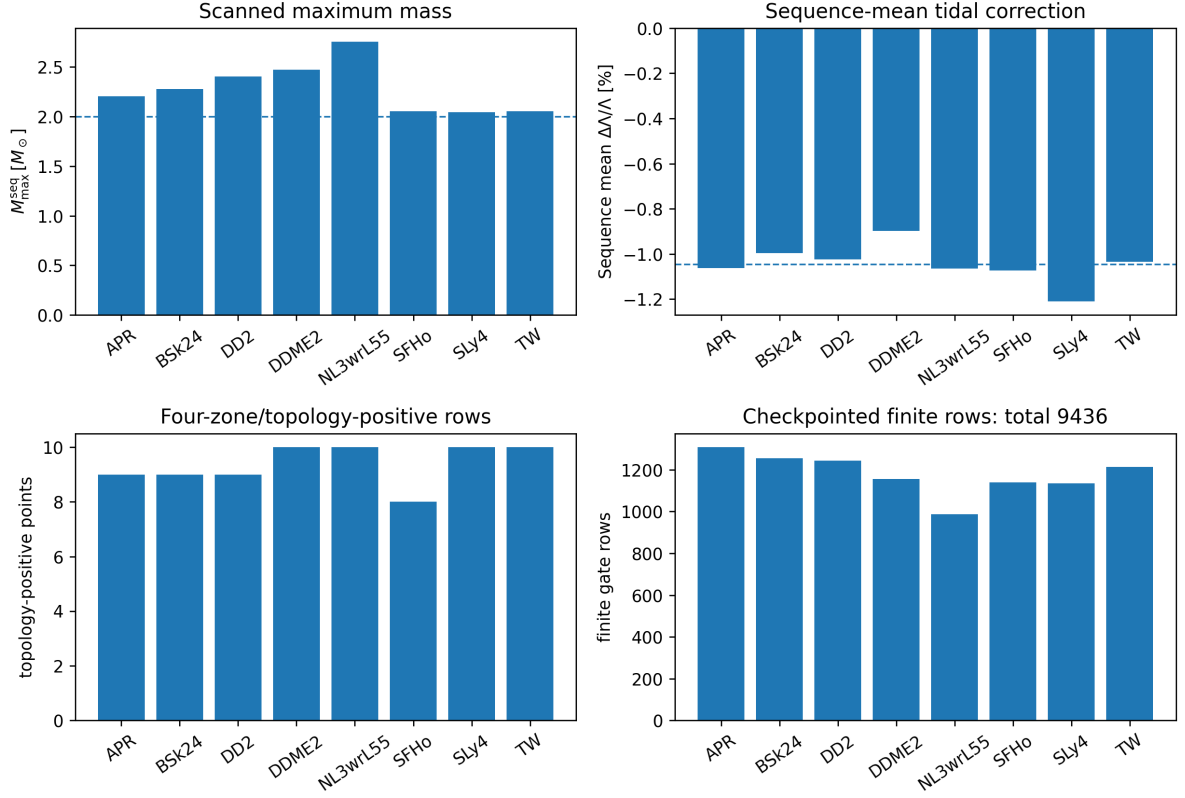


Figure 2: Strict global checkpoint gate. Top left: scanned maximum mass, with the $2M_{\odot}$ line shown for orientation. Top right: sequence-mean tidal correction. Bottom panels: topology-positive points and finite checkpoint rows.

The mass-dependent curves in Fig. 3 show that the suppression is not a single-table artifact. The correction remains negative throughout the exported sequence rows. The canonical mass $1.4M_{\odot}$ is marked because it is the standard reference point for tidal-deformability comparisons.

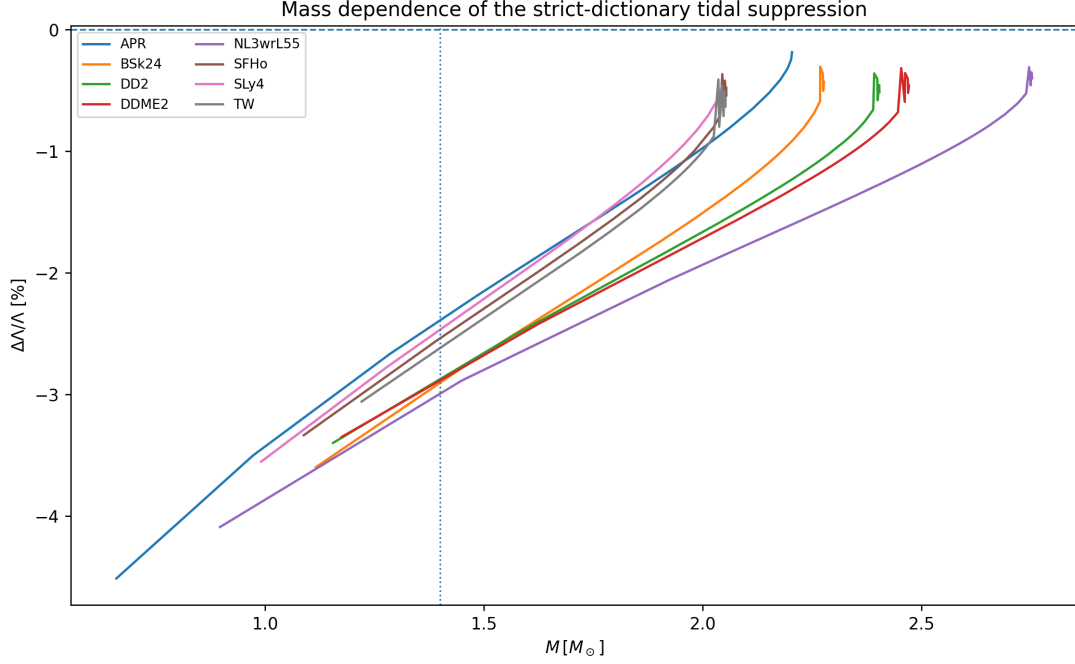


Figure 3: Mass dependence of the strict-dictionary tidal correction across all eight EOSs. The vertical dotted line marks $1.4M_{\odot}$. The correction is negative over the exported sequence rows, with a stronger suppression at lower masses and a smaller sequence average over the full scan.

6 Canonical $1.4M_{\odot}$ prediction

The refined interpolation at $1.4M_{\odot}$ is the cleanest observational summary. The audit-confirmed source of truth is the fine-interpolation column `deltaLambda_percent_1p4`. Table 2 reports the values used in the paper.

Table 2: Canonical-mass tidal prediction at $1.4M_{\odot}$. The last column uses the fine-interpolation source-of-truth estimator.

EOS	$R_{1.4}$ [km]	$\Lambda_{\text{GR},1.4}$	$\Lambda_{\text{BPB},1.4}$	$\Delta\Lambda/\Lambda(1.4M_{\odot})$ [%]
APR	10.280528	133.315225	130.121004	-2.385
BSk24	12.104734	372.668245	361.683802	-2.898
DD2	11.849238	339.796133	329.609574	-2.863
DDME2	11.859872	366.511621	355.373747	-2.871
NL3wrL55	12.295738	417.032716	404.445679	-2.982
SFHo	10.822294	160.358715	156.282492	-2.537
SLy4	10.547343	146.185774	142.543554	-2.463
TW	11.041789	196.750560	191.483257	-2.617
Mean	—	—	—	-2.702
Scatter	—	—	—	0.213

The result is compact:

$$\langle \Delta\Lambda/\Lambda \rangle_{1.4} = -2.7019\%, \quad \sigma_{\text{EOS}} = 0.213 \text{ percentage point}, \quad (10)$$

with range

$$-2.982\% \leq \Delta\Lambda/\Lambda(1.4M_{\odot}) \leq -2.385\%. \quad (11)$$

All eight EOSs are negative.

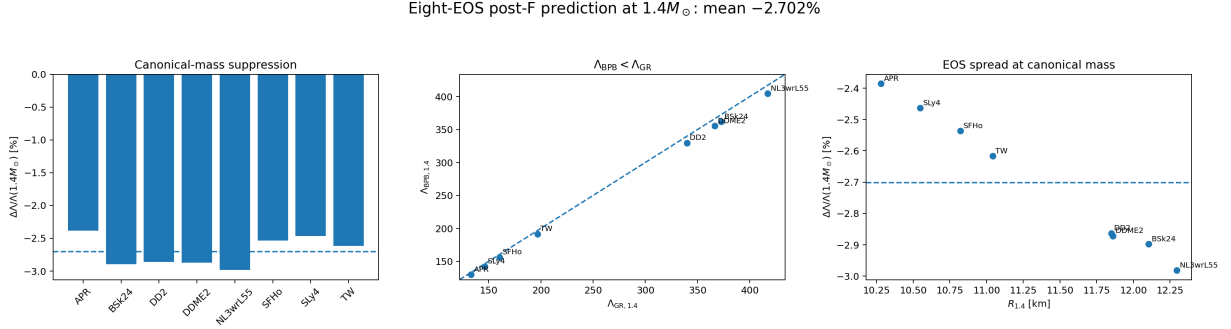


Figure 4: Canonical $1.4M_{\odot}$ prediction. Left: fine-interpolation fractional suppression. Middle: displayed Λ_{BPB} values lie below the $\Lambda_{BPB} = \Lambda_{GR}$ line for all EOSs. Right: the EOS spread in $R_{1.4}$ does not erase the common negative sign.

7 Response diagnostics

Figure 5 shows two row-level diagnostics from the exported sequences. The left panel displays the correction as a function of compactness proxy $2M/R$. The right panel displays the correction as a function of the local effective response α_{eff} . These panels should not be overinterpreted as a final analytic derivation, but they show that the exported suppression varies smoothly across the sequence rows and is tied to the effective local response rather than to a single isolated EOS point.

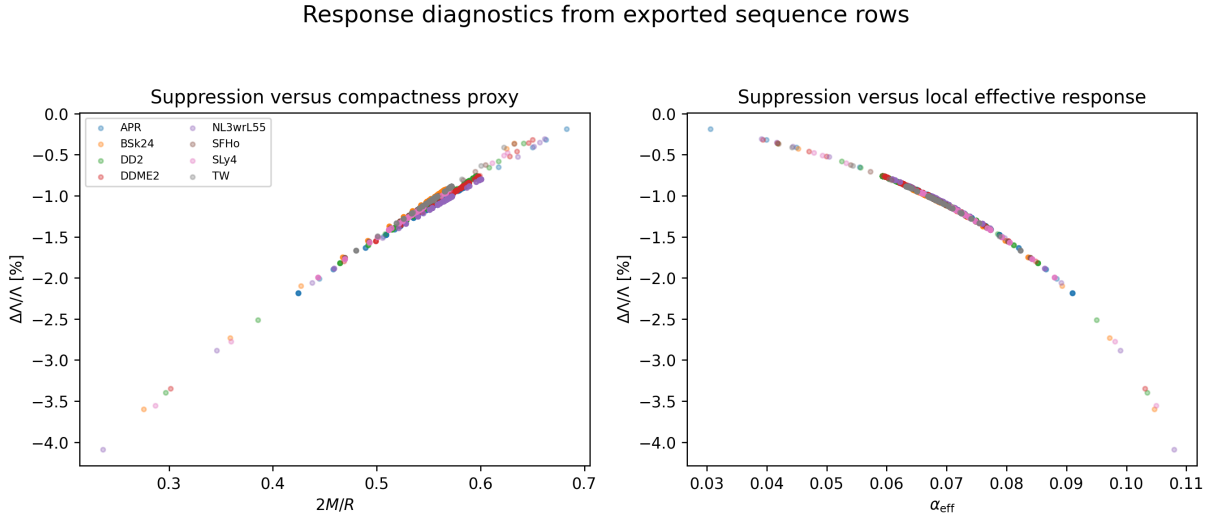


Figure 5: Response diagnostics from the exported strict sequence rows. The present outputs support an effective tidal/Love-sector reading of the correction. A full $R/C/k_2$ structural decomposition requires additional paired GR/BPB sequence outputs.

8 NS-3 output-supported decomposition

The NS-3 diagnostic was run after v0.2 to test whether the current strict outputs already support an independent decomposition into radius, compactness, and Love-number factors. The answer is precise: they do not yet contain separate GR/BPB radius or compactness columns. Consequently,

the audit-safe decomposition in the current outputs is

$$\frac{\Lambda_{\text{BPB}}}{\Lambda_{\text{GR}}} = \frac{k_{2,\text{eff}}^{\text{BPB}}}{k_{2,\text{GR}}} \left(\frac{C_{\text{GR}}}{C_{\text{BPB}}} \right)^5, \quad \left(\frac{C_{\text{GR}}}{C_{\text{BPB}}} \right)^5 = 1 \quad (12)$$

within the exported row-level diagnostic. Equivalently,

$$\Delta \ln \Lambda = \Delta \ln k_{2,\text{eff}}, \quad (13)$$

for the current checkpoint files. This is not a physical assertion that radius and compactness effects are absent in the underlying stellar problem. It is an output-level statement: the files currently available evaluate the GR and BPB tidal quantities on the same exported mass-radius support.

At the sequence level, NS-3 reproduces the global gate values:

$$\langle \Delta \Lambda / \Lambda \rangle_{\text{seq}} = -1.045625\%, \quad \sigma_{\text{EOS}} = 0.081708 \text{ pp}, \quad (14)$$

with 9436 rows and eight EOSs. At $1.4M_{\odot}$, NS-3 gives

$$\langle \Delta \Lambda / \Lambda \rangle_{1.4} = -2.701932\%, \quad \sigma_{1.4} = 0.212962 \text{ pp}, \quad (15)$$

with range $[-2.981504\%, -2.385498\%]$.

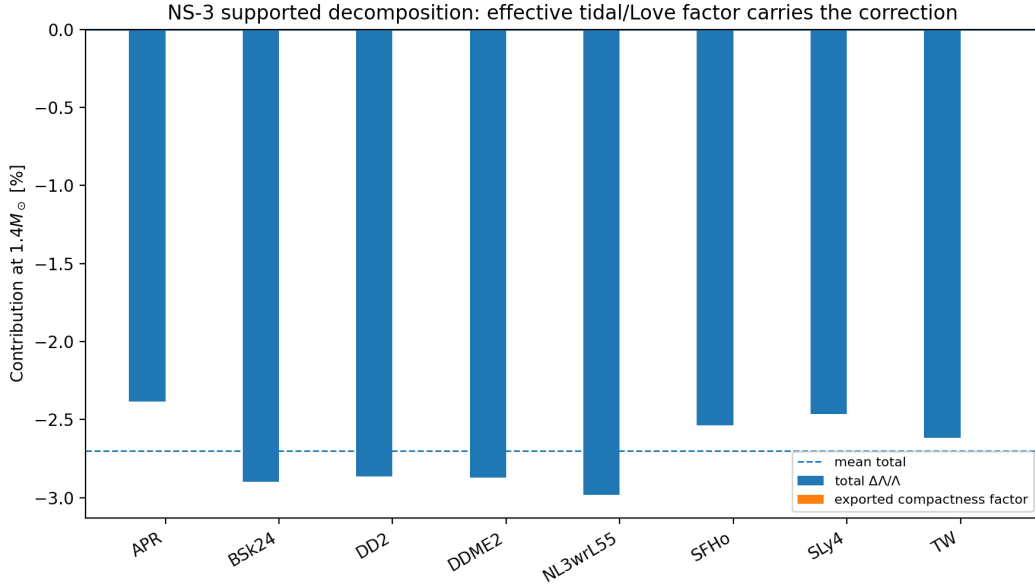


Figure 6: NS-3 canonical-mass decomposition at $1.4M_{\odot}$. Because the current strict outputs do not export separate GR/BPB radii or compactnesses, the compactness factor is unity and the percent-level suppression is carried by the effective Love/tidal factor.

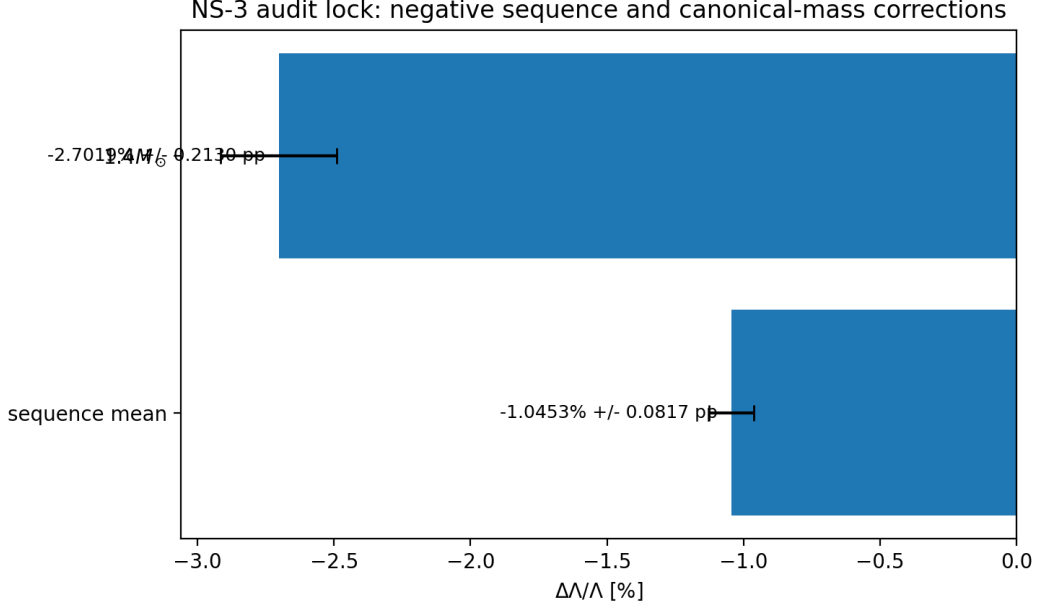


Figure 7: NS-3 sequence-mean decomposition. The mean sequence correction is reproduced as an effective k_2 response in the exported variables, with zero independent compactness/radius contribution in the current files.

Table 3 gives the canonical-mass Love-sector quantities extracted by NS-3. The important point is that the effective k_2 suppression has exactly the same percent correction as the total tidal suppression in the current outputs.

Table 3: NS-3 canonical-mass output-supported decomposition. The compactness factor is unity in the current exported rows, so $\Delta k_{2,\text{eff}}/k_{2,\text{GR}} = \Delta\Lambda/\Lambda$.

EOS	C_{export}	Λ_{GR}	Λ_{BPB}	$k_{2,\text{GR}}$	$k_{2,\text{eff}}^{\text{BPB}}$
APR	0.201086	133.315	130.121	0.065748	0.064173
BSk24	0.170782	372.668	361.684	0.081214	0.078820
DD2	0.174465	339.796	329.610	0.082385	0.079915
DDME2	0.174308	366.512	355.374	0.088465	0.085776
NL3wrL55	0.168129	417.033	404.446	0.084039	0.081502
SFHo	0.191020	160.359	156.282	0.061176	0.059621
SLy4	0.196000	146.186	142.544	0.063427	0.061846
TW	0.187223	196.751	191.483	0.067889	0.066072

The v0.4 wording is therefore source-locked more cleanly than v0.3.1: the paper no longer merely says that a true structural split is missing. It proves, from the NS-3 post-processing output, which decomposition is actually supported by the current files.

9 Triad stability

Table 4 and Fig. 8 summarise the local triad diagnostic. The strict branch remains the selected branch, but the nearby branches preserve the sign and order of magnitude. This is useful because it shows that the compact-object result is not a knife-edge effect of κ .

Table 4: Local dictionary triad. This is a stability diagnostic, not the primary result.

Branch	κ	mean $\Delta\Lambda/\Lambda$ [%]	$\alpha_{\text{eff}}/\alpha_{\infty}$
strict_dictionary	0.965543842	-0.769	0.380
legacy_PaperVI	0.997500000	-0.805	0.389
hybrid_working	1.072411000	-0.906	0.413

Dictionary stability: strict branch remains the selected branch

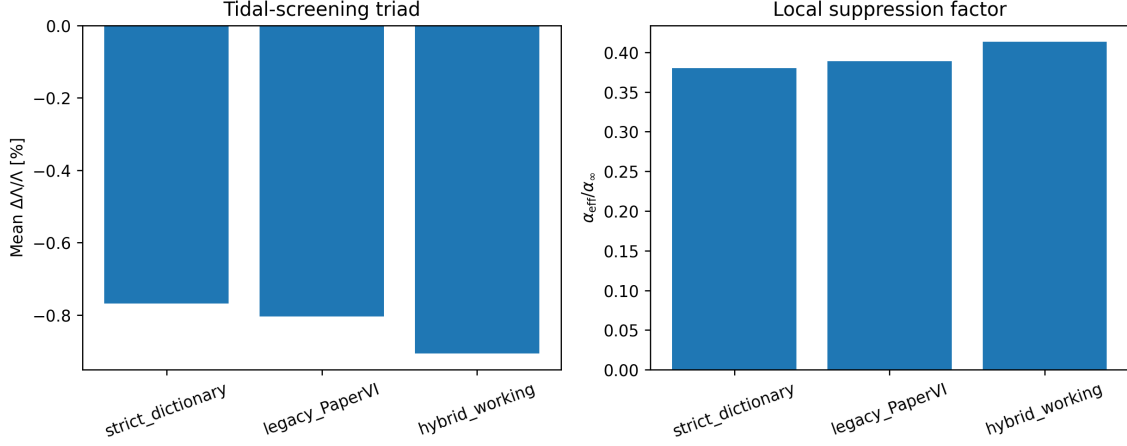


Figure 8: Triad stability. Nearby dictionary branches preserve the negative sign and the percent-level scale. The strict dictionary is retained as the main Paper IV branch because it is selected upstream by the CMB/Weyl gate.

10 Audit status

The v0.0 audit produced twelve initial flags, all resolved as false positives. The issue was the distinction between two columns:

```
deltaLambda_percent_1p4           # fine-interpolation source of truth
deltaLambda_percent_1p4_from_Lambdas # recomputed from displayed Lambdas
```

The paper uses the first column. Recomputing from the displayed Λ values gives slightly different values because it uses a different estimator and rounded/interpolated displayed quantities. The sign and the conclusion are unchanged.

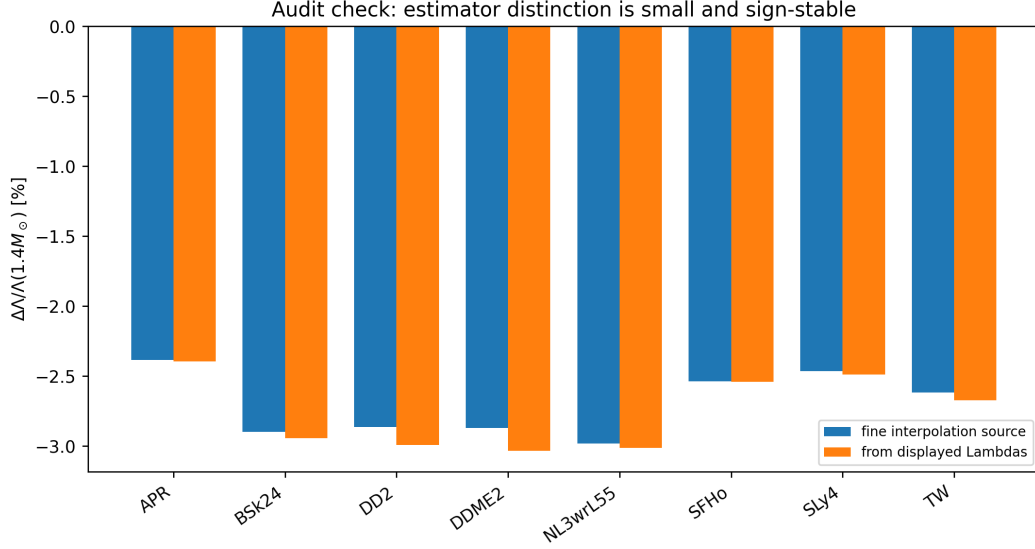


Figure 9: Audit check at $1.4M_{\odot}$. The fine-interpolation estimator is the paper source of truth. The displayed-Lambda recomputation is close and sign-stable, but not the value quoted for the locked statistics.

The locked audit numbers are:

$$\langle \Delta\Lambda/\Lambda \rangle_{1.4} = -2.7019\%, \quad \sigma_{1.4} = 0.213 \text{ pp}, \quad (16)$$

$$\langle \Delta\Lambda/\Lambda \rangle_{\text{seq}} = -1.0456\%, \quad \sigma_{\text{seq}} = 0.082 \text{ pp}. \quad (17)$$

Final audit verdict for v0.4.1: all eight sequence means are negative, all eight canonical-mass corrections are negative, and the quoted summary statistics match the source CSV values.

11 Relation to current gravitational-wave constraints

The compact-object gate should not be oversold. A -2.7% canonical-mass suppression is not a current event-level exclusion or detection claim. Present binary-neutron-star constraints, beginning with GW170817, still depend strongly on EOS priors, waveform assumptions, mass ratio, spin priors, and population assumptions [13]. Paper IV therefore claims only the following:

1. the strict branch remains viable in neutron-star structure;
2. it predicts a negative tidal residual at fixed EOS and mass;
3. the effect is small enough to avoid an immediate conflict with present uncertainties;
4. the sign and EOS coherence make it a future target for high-precision BNS inference.

This is the right level of ambition for the current outputs.

12 What v0.4 improves over v0.3.1

Version v0.4.1 makes the Paper-IV source consistent with the current corpus release.

1. It makes the paper corpus-aware and links it to the release-level files `00_README_CORPUS`, `00_MANIFEST_CORPUS`, `00_CLAIMS_INDEX`, and `00_REVIEWER_ROUTE`.

2. It records the post-F no-minus tidal-convention source lock and updates the canonical $1.4M_{\odot}$ Lambda table to the post-F values.
3. It keeps the key physics conclusion unchanged: all eight EOSs remain negative, the mean canonical suppression is -2.701932% , and the sequence-mean suppression remains about -1.046% .
4. It sharpens the guardrail: the present claim is an exported strict-dictionary effective tidal response, not a completed native scalar-tensor Love-number solver and not a full gravitational-wave population likelihood.

The NS-3 output-supported decomposition remains the audit-safe decomposition for the current files. A future matched-sequence export can still upgrade this to a native $R/C/k_2$ split.

13 Conclusions

The strict BPB/MBE branch selected by the previous gates survives the compact-object test. The numerical result is now sharp and audit-locked:

1. Eight EOSs are tested: APR, BSk24, DD2, DDME2, NL3wrL55, SFHo, SLy4, and TW.
2. The strict global checkpoint gate completes for all eight EOSs.
3. The finite sequence-level statistics use 9436 rows and include 75 topology-positive points.
4. All eight sequence-mean tidal corrections are negative.
5. The sequence-mean correction is -1.0456% with 0.082 pp EOS scatter.
6. At $1.4M_{\odot}$, all eight EOSs are negative.
7. The canonical-mass correction is -2.7019% with 0.213 pp EOS scatter.
8. The current output-level decomposition identifies the result as an effective tidal/Love-sector suppression at fixed compactness; a native full $R/C/k_2$ decomposition remains the next targeted upgrade.

Thus Paper IV establishes the compact-object gate of the BPB/MBE programme:

low-redshift growth, galactic dynamics, CMB/Weyl response, and neutron-star tidal structure all select a coherent strict branch, with the neutron-star sector adding a small but signed EOS-coherent tidal suppression.

A Reproducibility commands

Main strict gate:

```
python runs/ns/watch_strict_global_gate.py
```

Canonical-mass refinement:

```
python runs/ns/refine_1p4_from_strict_json_v2.py
```

Core source files:

```

outputs/ns/run_multieos8_strict_global_checkpoint_gate_summary.csv
outputs/ns/run_multieos8_strict_global_checkpoint_gate_rows.csv
outputs/ns/ns2_refine_1p4_from_strict_json_v2.csv
outputs/ns/run_multieos8_kappa_triad_tidal_screening_summary.csv
outputs/ns/ns_split_relic_dictionary_closure.json
outputs/ns/ns_strict_global_gate_freeze.md
outputs/ns/ns_strict_global_gate_freeze.json
outputs/ns/ns_paperIV_1p4_postF_source_lock.md

```

Files generated or carried forward for this v0.4.1 source lock:

```

paperIV_v0_4_sequence_summary_used.csv
paperIV_v0_4_refine_1p4_postF_used.csv
paperIV_v0_4_triad_summary_used.csv
paperIV_v0_4_stats_used.csv
ns3_decompose_strict_tidal_response_by_eos.csv
ns3_decompose_strict_tidal_response_1p4.csv
ns3_decompose_strict_tidal_response_rows.csv
ns3_decompose_strict_tidal_response.md

```

B Next targeted task

The NS-3 post-processing pass closed the decomposition question for the current outputs. The next target is not another global gate. It is a matched-sequence export that writes native paired quantities,

EOS, M , R_{GR} , R_{BPB} , C_{GR} , C_{BPB} , $k2_{GR}$, $k2_{BPB}$,
 Λ_{GR} , Λ_{BPB} , $d\ln R$, $d\ln C$, $d\ln k2$, $d\ln \Lambda$

at least at $1.4M_{\odot}$ and preferably over a compact mass grid. This will allow the physical decomposition

$$\Delta \ln \Lambda = \Delta \ln k_2 - 5\Delta \ln C \quad (18)$$

to replace the present output-supported effective Love-sector decomposition.

C Release-index files included in the source bundle

The source-only release bundle includes a `release_index/` directory containing the corpus-level navigation and source-lock indices used for the HAL corpus release:

```

release_index/00_MANIFEST_CORPUS.csv
release_index/00_CLAIMS_INDEX.csv
release_index/00_REVIEWER_ROUTE.md
release_index/00_RELEASE_NOTES.md

```

These files are not required to compile the paper. They are included so that a reviewer can navigate from the Paper-IV claims to the corpus-level release table.

References

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